

Middleware Architecture (IS3108/SCS3203)

Scenario-Based Class Assessment - 2023

Duration - 2 Hours

Assignment Scenario - SriBank PLC

SriBank PLC is (SriBank) a Sri Lankan-based commercial bank that is into both corporate and retail banking products and services. The bank has operated for over 20 years with more than 100 branches across the island. SriBank is considered one of the most technologically superior customer-centric organisations in the country.

SriBank implemented the core banking solution during the initial years and is currently, connected to all its branches in an online manner. The bank had also implemented a 500 ATM (Automated Teller Machines) network to extend its reach to its retail customers. The customers are also given the facility to use credit and debit cards certified by MASTER and VISA international payment networks.

The management of SriBank has strategised to implement an Internet and Mobile Banking solution with the objective of enhancing customer access to the bank's services through digital channels. The key functionalities expected out of the proposed Internet and Mobile Banking Platform are as follows.

1. The types of primary transactions that should be accommodated by this system are balance inquiry, account summary, funds transfer, utility bill payments, online account creation/sign-up, etc.
2. The solution should accommodate the needs of only retail banking customers (Corporate Banking should be considered out of scope). Retail customers should be properly authenticated (two-factor) for access to the platform.
3. The special service requests of the new Internet and Mobile Banking Platform are the ability to create new accounts, request loans and request special service approvals like credit limit increments, etc. However, these types of workflow-based transactions may not be online and real-time when compared to other types such as balance inquiry, funds transfer, etc. These special types of transactions may need workflow approvals from several departments/personnel that interact through different systems such as credit management, core banking, etc.
4. The Internet and Mobile Banking Platform should integrate with the following systems to provide overall functionality.
 - a. Core Banking System - maintains the customer profiles (KYC) and account information (balance, transaction history, interest, etc). The integration is based on RESTful services.

- b. Inter-bank Payment Gateway (e.g. CEFT, JustPay) - Provided by the Central Bank of Sri Lanka for routing payment requests across participating banks in Sri Lanka. The integration is based on ISO 8583 specifications.
- c. Credit Card System - The credit card information of customers such as credit card limits, balance, active status, billing cycle, minimum payment, etc, are maintained. The integration is based on SOAP standards.
- d. Internal Payment Switch - The payment switch links external international payment processing providers such as MasterCard and VISA. The integration is based on TCP/IP-based POX (Plain Old XML) interface.

The new platform for providing Internet and Mobile Banking should integrate with the above-mentioned systems. Mobile Banking should enable the use of iOS and Android platforms and Internet Banking should work on standard Internet browsers and the interfaces should be mobile responsive. The business logic for both mobile and internet banking should be common and amalgamated. The customer and other information should be persisted in a relational database. The mobile banking solution should provide notifications to the user's mobile application as push notifications as well as in the mode of SMS.

Exercise

1. Propose an architecture to realise the Internet and Mobile Banking Platform for the scenario given above.
 - a. Identify the layers and tiers of the architecture.
 - b. List the components of each layer and tier.
 - c. Sketch/draw the overall architecture with clear labels for tiers and components.
 - d. Discuss at least 3 architectural patterns you have used in the overall architecture of the solution.
2. Identify and describe a suitable architectural pattern for implementing the browser-based Internet banking scope of the solution. It is recommended to use an architecture that gives responsibility to the client tier to perform validations and verification of user inputs and better user responsiveness and experience.
3. As described above, there are 4 existing systems to integrate to realise the overall required functionality of the proposed Internet and Mobile Banking Platform. Propose the most suitable architectural pattern to integrate the 4 heterogeneous interfaces described in the above scenario. Further, list and described 3 suitable integration patterns available in the proposed architectural pattern that would be useful to integrate the external systems - core bank, Inter-Bank Payment Gateway, Credit Card System and the Internal Payment Switch.
4. Describe the nature of communication (synchronous/asynchronous) between the banking platform and the mobile applications with reference to sending notifications. Suggest the most suitable architectural pattern to implement the notification functionality.
5. The accuracy of the transactions is of prime importance to an Internet and Mobile banking Platform. The main concern is managing transaction concurrency whilst maintaining ACID properties. Propose and justify a tier/layer of the architecture that is most suitable for managing transaction concurrency.

Note: Any assumptions taken should be clearly mentioned.